Cell Chemistry and Bioenergetics

The Chemical Components of a Cell

Cells are composed of a limited number of elements and are governed by the laws of chemistry and physics.

Chemical Bonds

- **Atoms and Elements:** The cell is 96% carbon (C), hydrogen (H), nitrogen (N), and oxygen (O). These elements form the basis of organic chemistry.
- **Covalent Bonds:** Formed when two atoms share electrons. They are strong and stable in biological systems. The number of covalent bonds an atom can form is determined by its valence electrons (e.g., H:1, O:2, N:3, C:4).
- **Ionic Bonds:** Formed when electrons are donated from one atom to another, creating charged ions that are attracted to each other. They are weaker than covalent bonds in aqueous environments because water molecules can shield the ions.
- **Hydrogen Bonds:** A weak noncovalent interaction between a positively charged hydrogen atom (in a polar covalent bond, like O-H or N-H) and a negatively charged atom (like O or N). Crucial for the properties of water and the structure of macromolecules.
- **van der Waals Attractions:** Weak, short-range attractions between any two atoms, caused by fluctuating electrical charges. Significant when many atoms are close together.
- **Hydrophobic Force:** Not a true bond, but a force that pushes nonpolar surfaces out of the aqueous solution, driven by the tendency of water to form the maximum number of hydrogen bonds.
- **Figure 2-2 & 2-3:** Illustrate the polar nature of water and its ability to form hydrogen bonds.
- **Figure 2-4:** Compares covalent and ionic bonds.

Small Molecules in Cells

- The cell's small organic molecules (molecular weight 100-1000) are carbon-based compounds called organic molecules. They serve as building blocks (monomers) for larger macromolecules.
- **The Four Major Families:**
  1. **Sugars (Carbohydrates):**
     - **Monosaccharides:** Simple sugars with the general formula $(CH_2O)n$. Glucose ($C_6H_{12}O_6$) is a central energy source.
     - **Disaccharides:** Two monosaccharides linked by a glycosidic bond (e.g., sucrose = glucose + fructose).
     - **Oligosaccharides & Polysaccharides:** Chains of monosaccharides. Glycogen (in animals) and starch (in plants) are long-term stores of glucose. Cellulose forms plant cell walls.
     - **Figure 2-6:** Shows sugar structures and glycosidic bonds.
  2. **Fatty Acids (Lipids):**
     - **Structure:** A long hydrocarbon chain (hydrophobic) attached to a carboxyl (-COOH) group (hydrophilic). They are amphipathic.
     - **Saturated vs. Unsaturated:** Saturated fatty acids have no double bonds; unsaturated fatty acids have one or more double bonds, creating kinks in the chain.

    - **Function:** Concentrated energy storage (triacylglycerols) and the primary components of cell membranes (phospholipids).
    - **Figure 2-7:** Details the structure of fatty acids and their assembly into fats and membrane lipids.
3. **Amino Acids:**
    - **Structure:** A central α-carbon atom linked to an amino group ($-NH_2$), a carboxyl group (-COOH), a hydrogen atom, and a variable side chain (R group).
    - **Function:** The building blocks of proteins. The 20 common amino acids have different side chains, giving them unique chemical properties (e.g., acidic, basic, polar, nonpolar).
    - **Peptide Bonds:** Amino acids are joined by peptide bonds to form polypeptide chains.
    - **Figure 2-8 & 2-9:** Show the 20 amino acids and the formation of a peptide bond.
4. **Nucleotides:**
    - **Structure:** A nitrogen-containing base, a five-carbon sugar (ribose or deoxyribose), and one or more phosphate groups.
    - **Bases:** Pyrimidines (Cytosine, Thymine, Uracil) and Purines (Guanine, Adenine).
    - **Function:** Building blocks of nucleic acids (DNA and RNA).
    - **ATP (Adenosine Triphosphate):** A key nucleotide that serves as the principal short-term carrier of chemical energy in the cell. Its high-energy phosphoanhydride bonds release energy when hydrolyzed.
    - **Phosphodiester Bonds:** Nucleotides are joined by phosphodiester bonds to form nucleic acid chains.
    - **Figure 2-10 & 2-11:** Illustrate nucleotide structure and the role of ATP.

Macromolecules in Cells

- **Polymers:** Long chains of monomers linked by covalent bonds.
- **Structure and Function:** The specific sequence of monomers determines the unique three-dimensional structure and biological function of a macromolecule.
- **Noncovalent Bonds in Macromolecules:** The folded conformation of a macromolecule is stabilized by a large number of weak noncovalent bonds (hydrogen bonds, ionic bonds, van der Waals attractions, and hydrophobic interactions).
- **Self-Assembly:** The unique conformation is typically the one that minimizes free energy.
- **Figure 2-12 & 2-13:** Show the noncovalent interactions that determine macromolecular shape.
- **Synthesis:** Macromolecules are synthesized in condensation reactions, which are energetically unfavorable and require an input of energy (often from ATP hydrolysis).
- **Figure 2-14:** Depicts the synthesis of macromolecules via condensation reactions.

Catalysis and the Use of Energy by Cells

Cells obey the laws of thermodynamics, creating and maintaining order by capturing and using energy from their environment.

The Laws of Thermodynamics

- **First Law:** Energy cannot be created or destroyed, only converted from one form to another. (Conservation of Energy).
- **Second Law:** In any isolated system, the degree of disorder (entropy) can only increase.

- **Cells and Entropy:** Cells are not isolated systems. They use energy from their environment (e.g., sunlight, food) to create local order (biosynthesis), but in doing so, they release heat, which increases the total entropy of the universe.
- **Figure 2-15:** Illustrates the first and second laws of thermodynamics.

Photosynthesis and Cellular Respiration

- **Photosynthesis:** The process by which plants, algae, and some bacteria use sunlight to synthesize energy-rich organic molecules (sugars) from CO_2 and H_2O.
- **Cellular Respiration:** The process by which cells break down food molecules, releasing stored chemical energy and capturing it in the form of ATP.
- **Complementary Processes:** The O_2 released during photosynthesis is consumed during cellular respiration, and the CO_2 released during respiration is used for photosynthesis.
- **Figure 2-16:** Shows the complementary cycle of photosynthesis and cellular respiration.

Free Energy and Catalysis

- **Free Energy (G):** The energy in a system that is available to do work. Chemical reactions proceed in the direction that leads to a loss of free energy.
- **Free-Energy Change (ΔG):**
  - **ΔG < 0 (Negative):** The reaction is energetically favorable (spontaneous, exergonic). It releases energy.
  - **ΔG > 0 (Positive):** The reaction is energetically unfavorable (non-spontaneous, endergonic). It requires an input of energy.
  - **ΔG = 0:** The reaction is at equilibrium.
- **Standard Free-Energy Change (ΔG°):** The ΔG of a reaction under standard conditions (1M concentration for all reactants and products). It allows for the comparison of different reactions.
- **Equilibrium Constant (K):** $K = \frac{[X]}{[Y]}$ for the reaction Y → X at equilibrium. The relationship between ΔG° and K is given by: $\Delta G° = -RTlnK$.
- **Figure 2-17 & 2-18:** Explain the concept of free energy and its relationship to reaction spontaneity.

Enzymes and Activation Energy

- **Enzymes:** Highly specific protein catalysts that speed up chemical reactions without being changed themselves.
- **Activation Energy:** The energy barrier that must be overcome for a reaction to occur.
- **Mechanism:** Enzymes bind to specific substrate molecules and lower the activation energy of the reaction, thereby increasing the reaction rate. They do not change the ΔG or the equilibrium point of the reaction.
- **Figure 2-22 & 2-23:** Show how an enzyme lowers activation energy and illustrate a typical catalytic cycle.

Activated Carrier Molecules and Biosynthesis

- **Reaction Coupling:** Cells drive energetically unfavorable reactions (e.g., biosynthesis) by coupling them to energetically favorable ones, most often the hydrolysis of ATP.
- **Activated Carriers:** Small organic molecules that store and transfer energy in a readily usable form.

- **ATP:** The most important and versatile activated carrier. It carries energy in its high-energy phosphate bonds.
- **NADH and NADPH:** Carry high-energy electrons (reducing power). NADH is primarily involved in catabolism (breaking down food), while NADPH is primarily used in anabolism (biosynthesis).
- **Acetyl CoA:** Carries an acetyl group, which is used in the citric acid cycle and in building larger molecules.

- **Figure 2-19 & 2-20:** Illustrate the principle of reaction coupling and the role of ATP.
- **Figure 2-27:** Shows the major activated carriers used in metabolism.

How Cells Obtain Energy from Food

Cells break down food molecules in a stepwise manner to efficiently capture energy in the form of ATP and other activated carriers.

The Breakdown and Utilization of Sugars and Fats

- **Catabolism:** The set of metabolic pathways that breaks down food molecules into smaller units, releasing energy.
- **Anabolism (Biosynthesis):** The set of metabolic pathways that construct molecules from smaller units, requiring energy.
- **Figure 2-29:** Provides an overview of the relationship between catabolism and anabolism.

The Three Stages of Catabolism

1. **Stage 1: Digestion**
   - Breakdown of large macromolecules (proteins, polysaccharides, fats) into their monomer subunits (amino acids, sugars, fatty acids).
   - Occurs mostly outside cells (e.g., in the intestine) or in lysosomes.
2. **Stage 2: Glycolysis and Pyruvate Oxidation**
   - A chain of reactions called **glycolysis** splits each molecule of glucose into two smaller molecules of pyruvate.
   - Occurs in the cytosol.
   - Generates a small amount of ATP and NADH.
   - Pyruvate is then transported into the mitochondria and converted into acetyl CoA.
3. **Stage 3: Citric Acid Cycle and Oxidative Phosphorylation**
   - Occurs entirely in the mitochondria.
   - **Citric Acid Cycle (Krebs Cycle):** The acetyl group from acetyl CoA is completely oxidized to CO_2. Large amounts of NADH are generated.
   - **Oxidative Phosphorylation:** The high-energy electrons from NADH are passed along an **electron-transport chain** embedded in the inner mitochondrial membrane. The energy released is used to pump protons, creating a proton gradient that drives the synthesis of large amounts of ATP. This process consumes O_2.

- **Figure 2-30:** A diagram summarizing the three stages of food breakdown.
- **Figure 2-31, 2-32, 2-33:** Detail the processes of glycolysis, the citric acid cycle, and oxidative phosphorylation.

Storing and Utilizing Food

- **Gluconeogenesis:** An anabolic pathway that synthesizes glucose from small organic molecules like pyruvate or lactate. It is essentially the reverse of glycolysis but uses special enzymes to bypass the irreversible steps.
- **Food Storage:**
  - **Glycogen:** Animals store excess glucose as the branched polysaccharide glycogen, mainly in liver and muscle cells.
  - **Starch:** Plants store excess glucose as starch.
  - **Fats (Triacylglycerols):** Both plants and animals store excess energy much more efficiently as fats. Fats yield more than twice as much energy per gram as carbohydrates.
- **Metabolic Balance:** Cells continuously adjust their metabolic pathways, breaking down or synthesizing glycogen and fats to meet their immediate energy needs.
- **Figure 2-35, 2-36, 2-37:** Illustrate gluconeogenesis and the storage of sugars and fats.